Notes: Midrigan & Xu (AER, 2014)

Finance and Misallocation: Evidence from Plant-Level Data

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1 Big picture

- **Question:** How much can *financial frictions* (limited external finance / collateral constraints) explain aggregate *TFP losses from misallocation* (dispersion in marginal products across plants)?

- **Core message:** In plant data, "measured misallocation" can be large, but a quantitatively disciplined finance-only model produces *small* steady-state TFP losses from within-sector capital misallocation.

- **Main intuition:**

  1. A collateral constraint creates a plant-specific wedge in the shadow cost of capital,
  2. but productive plants save, accumulate net worth, and *grow out of* the constraint,
  3. so wedge dispersion is limited when productivity shocks are of empirically realistic size/persistence.

  Finance matters more through **extensive margins** (entry, adoption, operating/exit) than through persistent within-sector misallocation.

2 Benchmark model (two sectors + collateral constraint + equity issuance)

2.1 Agents and shocks

- Time $t = 0, 1, 2, \ldots$.

- A continuum of **producers** (plants). Each plant has permanent productivity z and transitory productivity e_t.

- Transitory productivity follows a finite-state Markov chain with transition matrix f_{ij}.

- A representative **worker** supplies labor inelastically and can save in a risk-free asset and hold plant equity claims.

- Preferences are log: $\sum_{t\geq 0} \beta^t \log C_t$.

2.2 Traditional sector

Traditional producers use labor only:

$$Y_t = \exp(z + e_t)^{1-\eta} L_t^{\eta}, \qquad \eta \in (0, 1). \tag{1}$$

They cannot borrow (only save):

$$C_t = Y_t - W_t L_t - (1 + r)D_t + D_{t+1}, \qquad D_{t+1} \leq 0. \tag{2}$$

2.3 Modern sector: entry cost, production, and finance

Entry cost. To operate the modern technology, a plant pays a sunk cost $\exp(z)\kappa$.

Collateral constraint.

$$D_{t+1} \leq \theta\big(K_{t+1} + \exp(z)\kappa\big), \qquad \theta \in [0, 1). \tag{3}$$

Equity issuance at entry. Upon entry, the plant can sell a fraction $\theta\chi$ of the equity claim to its future profits. Let P_t be the value of the entire profit stream. Entry proceeds include $\theta\chi P_t$.

Entry-period budget constraint.

$$C_t + K_{t+1} + \exp(z)\kappa = Y_t - W_t L_t - (1 + r)D_t + D_{t+1} + \theta\chi P_t. \tag{4}$$

Modern production.

$$Y_t = \exp(z + e_t + \phi)^{1-\eta}\big(L_t^{\alpha} K_t^{1-\alpha}\big)^{\eta}.$$

Define operating profit (before equity payout):

$$\Pi_t^m \equiv Y_t - W_t L_t - (r + \delta)K_t.$$

Budget constraint after entry. Because the plant sold a share $\theta\chi$ of its profits at entry:

$$C_t + K_{t+1} - (1 - \delta)K_t = Y_t - W_t L_t - (1 + r)D_t - \theta\chi\Pi_t^m + D_{t+1}. \tag{5}$$

2.4 Recursive formulation (scaled net worth)

Define net worth $A_t \equiv K_t - D_t$ and scale by $\exp(z)$:

$$a_t \equiv \frac{A_t}{\exp(z)}, \qquad k_t \equiv \frac{K_t}{\exp(z)}.$$

Homogeneity allows the modern producer problem to be written in (a, e).

Modern value function.

$$V^m(a, e_i) = \max_{a', c} \left\{ \log c + \beta \sum_j f_{ij} V^m(a', e_j) \right\}. \tag{6}$$

Scaled flow budget:

$$c + a' = (1 - \theta\chi)\pi^m(a, e) + (1 + r)a. \tag{7}$$

Scaled operating profit:

$$\pi^m(a, e) = \max_{k,l} \left\{ \exp(e + \phi)^{1-\eta}\big(l^{\alpha} k^{1-\alpha}\big)^{\eta} - Wl - (r + \delta)k \right\}. \tag{8}$$

Collateral constraint in scaled variables. Using $D = K - A$ in (3) implies:

$$k \leq \frac{1}{1 - \theta} a + \frac{\theta}{1 - \theta}\kappa. \tag{9}$$

Euler equation with wedge. Let $\mu(a', e_j) \geq 0$ be the multiplier on (9). Then:

$$\frac{1}{c(a, e_i)} = \beta \sum_j f_{ij} \left[(1+r) + \frac{1}{1-\theta} \mu(a', e_j) \right] \frac{1}{c(a', e_j)}. \tag{10}$$

Intuition: if $\mu > 0$, additional net worth relaxes the constraint, raising the effective return to saving.

2.5 Static factor demands and the "shadow cost of funds"

FOC for labor:

$$\alpha\eta \, \frac{y(a, e)}{l(a, e)} = W. \tag{11}$$

FOC for capital:

$$(1-\alpha)\eta \, \frac{y(a, e)}{k(a, e)} = r + \delta + \mu(a, e). \tag{12}$$

So $\mu(a, e)$ is a plant-specific wedge in the shadow cost of capital.

3 Traditional producers and the entry decision

3.1 Traditional value function (unnumbered in the paper)

Traditional producers cannot borrow. Let $a \geq 0$ denote their (scaled) net worth. They choose consumption and saving, and then decide whether to remain traditional or enter modern next period. A convenient recursive form is:

$$V^\tau(a, e_i) = \max_{a', c} \left\{ \log c + \beta \max \left[\sum_j f_{ij} V^\tau(a', e_j), \ \sum_j f_{ij} V^m(a', e_j) \right] \right\}.$$

3.2 Traditional budget constraint

$$c + x = \pi^\tau(e) + (1+r)a, \qquad \pi^\tau(e) = \max_l \left\{ \exp(e)^{1-\eta} l^\eta - W l \right\}. \tag{13}$$

3.3 Net worth if entering modern + equity issuance

If the producer enters modern, it pays κ (scaled) but can raise funds by selling equity at entry:

$$a' = x - \kappa + \theta\chi \, p(a', e_i). \tag{14}$$

The (scaled) equity price satisfies:

$$p(a, e_i) = \frac{1}{1+r} \sum_j f_{ij} \left[p(a', e_j) + \pi^m(a', e_j) \right]. \tag{15}$$

3.4 Natural borrowing limit

$$a > a_{\min} = -\frac{(1-\theta\chi)\pi^m(a_{\min}, e_1)}{r}. \tag{16}$$

4 Equilibrium distributions and accounting

Let $n_t^m(A,e)$ and $n_t^\tau(A,e)$ be the time-t measures of modern and traditional producers. Their laws of motion are:

$$n_{t+1}^m(A,e_j) = \int_A \sum_i f_{ij}\,\mathbb{1}_{\{a^m(a,e_i)\in A\}}\ \mathrm{d}n_t^m(a,e_i) + \int_A \sum_i f_{ij}\,\mathbb{1}_{\{\xi(a,e_i)=1,\ a^{\tau,s}(a,e_i)\in A\}}\ \mathrm{d}n_t^\tau(a,e_i), \tag{17}$$

$$n_{t+1}^\tau(A,e_j) = \int_A \sum_i f_{ij}\,\mathbb{1}_{\{\xi(a,e_i)=0,\ a^{\tau}(a,e_i)\in A\}}\ \mathrm{d}n_t^\tau(a,e_i) + (\gamma-1)N_t\,\mathbb{1}_{\{0\in A\}}\ \bar{f}_j. \tag{18}$$

4.1 Asset market clearing

$$A_{t+1}^w + \sum_{i=m,\tau} \int_{A\times E} a_{t+1}^i(a,e)\ \mathrm{d}n_{t+1}^i(a,e) = \int_{A\times E} k_{t+1}^m(a,e)\ \mathrm{d}n_{t+1}^m(a,e). \tag{19}$$

4.2 Resource constraint

$$C_t + K_{t+1} - (1-\delta)K_t + X_t = Y_t. \tag{20}$$

5 Misallocation and TFP accounting (key results)

5.1 Actual modern-sector output and TFP

Using firm FOCs, modern-sector output can be written as:

$$Y = \exp(\phi)^{1-\eta}\,\frac{\left(\int_{i\in M}\exp(e_i)\,(r+\delta+\mu_i)^{-\frac{(1-\alpha)\eta}{1-\eta}}\ \mathrm{d}i\right)^{1-\alpha\eta}}{\left(\int_{i\in M}\exp(e_i)\,(r+\delta+\mu_i)^{\frac{\alpha\eta-1}{1-\eta}}\ \mathrm{d}i\right)^{(1-\alpha)\eta}}\left(L^\alpha K^{1-\alpha}\right)^\eta. \tag{21}$$

5.2 Efficient TFP holding fixed (K,L) and the set M

$$Y^e = \underbrace{\exp(\phi)^{1-\eta}\left(\int_{i\in M}\exp(e_i)\,\mathrm{d}i\right)^{1-\eta}}_{\text{TFP}^e}\left(L^\alpha K^{1-\alpha}\right)^\eta. \tag{22}$$

5.3 TFP loss from misallocation

$$\text{Loss} = \log\left(\int_{i\in M}\exp(e_i)\,\mathrm{d}i\right)^{1-\eta} - \log\left[\frac{\left(\int_{i\in M}\exp(e_i)\left(\frac{y_i}{k_i}\right)^{-\frac{(1-\alpha)\eta}{1-\eta}}\ \mathrm{d}i\right)^{1-\alpha\eta}}{\left(\int_{i\in M}\exp(e_i)\left(\frac{y_i}{k_i}\right)^{\frac{\alpha\eta-1}{1-\eta}}\ \mathrm{d}i\right)^{(1-\alpha)\eta}}\right]. \tag{23}$$

If $\log(y_i/k_i)$ and e_i are jointly normal:

$$\text{Loss} = \frac{1}{2}\,\frac{(1-\alpha\eta)(1-\alpha)\eta}{1-\eta}\ \mathrm{Var}\!\left(\log\frac{y_i}{k_i}\right). \tag{23$'$}$$

6 Planner benchmark (first best) and the extensive margin

The planner chooses measures n_i^τ, n_i^m, labor allocations (L^τ, L^m), and K:

$$\max_{\{n_i^\tau, n_i^m, L^\tau, L^m, K\}} \quad \left(\sum_i \exp(e_i) n_i^\tau\right)^{1-\eta} (L^\tau)^\eta + \left(\sum_i \exp(e_i+\phi) n_i^m\right)^{1-\eta} \left((L^m)^\alpha K^{1-\alpha}\right)^\eta$$
$$- \left(\delta + \frac{\gamma}{\beta} - 1\right) K - \sum_i \frac{n_i^m}{\beta}(\gamma - 1)\kappa. \tag{24}$$

Key point: finance frictions affect aggregate outcomes both through within-modern misallocation and through the *size of the modern sector* (entry/adoption/operation).

7 Extensions (additional "models" in the paper)

7.1 Model extension A: technology adoption within the modern sector

The paper adds a second (more productive) technology within the modern sector. A plant can incur an additional sunk cost $\exp(z)\kappa_p$ to gain an extra productivity shifter ϕ_p:

$$Y_t = \exp(z + e_t + \phi + \phi_p)^{1-\eta} \left(L_t^\alpha K_t^{1-\alpha}\right)^\eta.$$

They parameterize ϕ_p and set $\kappa_p = \exp(\phi_p)\kappa$, and then choose κ to match intangible investment.

Intuition: finance affects *adoption* of the productive technology (another extensive margin). In Table 6, eliminating external finance collapses the share adopting productive tech (from $\approx 86\%$ to $\approx 18\%$).

7.2 Model extension B: producer exit / operating decision

Modern producers face a fixed operating cost each period (proportional to permanent productivity). Each period, a modern producer chooses whether to pay the cost and operate modern or instead produce in the traditional technology. The budget constraint becomes:

$$c + a' = \max\left[\pi^m(a, e_i) - WF,\ \pi^\tau(e_i)\right] + (1+r)a,$$

where F is the fixed operating cost in labor units (so WF is the output cost).

Intuition: finance frictions can trigger inefficient shutdown/operation choices for low-net-worth plants, affecting the age distribution.

7.3 Model extension C: "no entry" economy (isolating the intensive margin)

To isolate within-sector capital misallocation, the paper studies an economy where all producers are in the modern sector (no entry margin). A useful upper bound on TFP losses (if capital is fixed) is:

$$\text{Loss} = (1-\eta)\log\left(\int_{i\in M} \exp(e_i)\,\mathrm{d}i\right) - (1-\alpha\eta)\log\left(\int_{i\in M} \exp(e_i)^{\frac{1-\eta}{1-\alpha\eta}}\,\mathrm{d}i\right) = \frac{1}{2}\frac{(1-\alpha)\eta}{1-\alpha\eta}(1-\eta)\sigma_e^2, \tag{25}$$

under lognormal e.

8 Empirical decomposition used in the paper

To separate an "age" channel from within-age dispersion, they run:

$$\log\left(\frac{y_i}{k_i}\right) = \sum_a \gamma_a D_{a,i} + \varepsilon_i, \tag{26}$$

where $D_{a,i}$ are age dummies. In the data, the between-age component is small relative to within-age dispersion.

9 Interpreting the tables and figures (with key numbers)

9.1 Figure 1 (policy functions): constraints bind only at low net worth

- **Panel A:** the wedge $\mu(a, e)$ is large when a is low and quickly falls toward 0 as a rises.
- **Panel B:** constrained plants save more (higher $a'(a, e)$), so constraints are self-relaxing over the life cycle.

9.2 Figure 2 (entry rule): z and a jointly determine entry

Higher z (permanent productivity) shifts the entry threshold down: high-z plants enter modern earlier (at lower net worth).

9.3 Table 1 (moments): disciplined matching of plant dynamics + finance aggregates

- The benchmark (and extensions) match output volatility/persistence and key financing ratios (debt and equity issuance).
- They also match life-cycle patterns: young plants grow faster and have higher average products of capital.

9.4 Table 2 (parameters): finance and productivity processes

Key calibrated objects:

- (θ, χ): collateral and equity issuance pinned down by debt/value added and equity issuance moments.
- $(\sigma_\varepsilon, \rho)$: transitory productivity volatility/persistence pinned down by output moments.
- κ: entry/intangible cost pinned down by intangible investment share.

9.5 Table 3 (open economy, varying θ): misallocation is modest; entry effects are large

With equity issuance (Panel A):

- Korea calibration: **misallocation loss** $\approx 0.3\%$ and **fraction modern producers** ≈ 0.93.

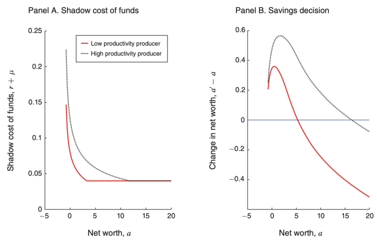

FIGURE 1. DECISION RULES: MODERN SECTOR

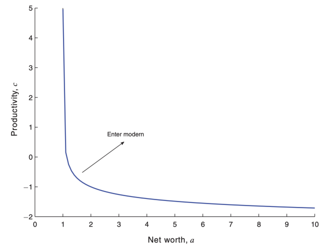

FIGURE 2. DECISION TO ENTER MODERN SECTOR

TABLE 1—MOMENTS

	Data Korea	Benchmark	Adoption	Exit
Panel A. Used to calibrate model				
SD output growth	0.59	0.58	0.58	0.59
SD output	1.31	1.30	1.37	1.30
1-year autocorrelation	0.90	0.90	0.89	0.91
3-year autocorrelation	0.87	0.87	0.83	0.87
5-year autocorrelation	0.85	0.86	0.80	0.86
Intangibles invest. to output, percent	4.6	4.6	4.6	4.6
Output growth rate, percent	8.0	8.0	8.0	8.0
Debt to output	1.2	1.2	1.2	1.2
Equity to output	0.3	0.3	0.3	0.3
Panel B. Additional moments				
SD employment growth	0.49	0.58	0.58	0.59
SD capital growth	0.57	0.57	0.53	0.52
SD employment	1.21	1.30	1.37	1.30
SD capital	1.44	1.30	1.40	1.30
1-year autocorrelation employment	0.92	0.90	0.89	0.91
5-year autocorrelation employment	0.86	0.86	0.80	0.86
1-year autocorrelation capital	0.92	0.90	0.92	0.92
5-year autocorrelation capital	0.86	0.86	0.79	0.88
Share producers, ages 1–5	0.51	0.32	0.32	0.64
Share producers, ages 6–10	0.27	0.22	0.22	0.12
Share output, ages 1–5	0.20	0.28	0.16	0.57
Share output, ages 6–10	0.20	0.23	0.23	0.12
Share employment, ages 1–5	0.21	0.28	0.16	0.57
Share employment, ages 6–10	0.20	0.23	0.23	0.12
Share capital, ages 1–5	0.22	0.26	0.13	0.55
Share capital, ages 6–10	0.21	0.23	0.23	0.12
Rel. output growth 1–5 vs. 11+	0.11	0.09	0.28	0.10
Rel. output growth 6–10 vs. 11+	0.05	0.01	0.08	0.04
Rel. employm. growth 1–5 vs. 11+	0.09	0.09	0.28	0.10
Rel. employm. growth 6–10 vs. 11+	0.02	0.01	0.08	0.04
Rel. capital growth 1–5 vs. 11+	0.09	0.13	0.27	0.12
Rel. capital growth 6–10 vs. 11+	−0.03	0.02	0.11	0.06

- No external finance ($\theta = 0$): misallocation rises to $\approx 4.7\%$, but the fraction modern producers collapses to ≈ 0.35; consumption falls to ≈ 0.82 (normalized).

Takeaway: the big welfare/output effect at low θ is mainly through the extensive margin (modern sector size), not within-modern misallocation.

9.6 Table 4 (closed economy): endogenous r can amplify effects at low θ

When the economy is closed, r adjusts. At very low θ, misallocation losses can be larger (e.g., $\approx 9.8\%$ at $\theta = 0$ in Panel A), but the baseline Korea-like loss remains small.

TABLE 2—PARAMETER VALUES

		Benchmark	Adoption	Exit
Assigned parameters				
Labor elasticity	α	0.67	0.67	0.67
Span of control	η	0.85	0.85	0.85
Capital depreciation	δ	0.06	0.06	0.06
Discount factor	$\beta(1+\mu)^{-1}$	0.92	0.92	0.92
Growth rate	γ	1.08	1.08	1.08
Persistence unit worker state	λ_1	0.79	0.79	0.79
Persistence zero worker state	λ_0	0.50	0.50	0.50
Relative efficiency in modern sector	$(1-\eta)\phi$	0.20	0.20	0.20
Calibrated parameters				
Collateral constraint	θ	0.86	0.78	0.68
Equity issuance constraint	χ	0.10	0.08	0
SD transitory shocks	σ_ε	0.50	0.50	0.96
Persistence transitory shocks	ρ	0.25	0.11	0.40
Cost of entering modern sector	κ	1.19	0.30	2.66
Variance exogenous permanent component	var(z)	1.47	1.43	1.44
Relative efficiency of productive technology	$(1-\eta)\phi_p$	—	0.27	—
Cost of adopting productive technology	κ_p	—	1.83	—
Fixed cost of operating in modern sector	F	—	—	0.27

TABLE 3—AGGREGATE IMPLICATIONS OF FINANCE FRICTIONS: OPEN ECONOMY

	Efficient	"Korea"	$\theta = 1$	$\theta = 0.75$	$\theta = 0.50$	$\theta = 0.25$	$\theta = 0$
Panel A. With equity issuance ($\chi = 0.10$)							
Debt to output (modern)		1.18	1.30	0.92	0.35	−0.14	−0.60
Equity to output (modern)		0.29	0.32	0.30	0.26	0.14	0
Interest rate		0.047	0.047	0.047	0.047	0.047	0.047
Fraction constrained		0.17	0	0.44	0.69	0.78	0.83
Capital to output (modern)	1.93	2.59	2.65	2.46	2.25	2.14	2.05
TFP (modern)	1.003	1.000	1.003	0.989	0.926	0.869	0.827
Loss misallocation, percent	0	0.3	0.0	1.4	3.9	4.4	4.7
Fraction producers modern	0.93	0.93	0.93	0.93	0.70	0.48	0.35
Fraction output modern	0.99	0.99	0.99	0.99	0.90	0.73	0.58
Consumption	1.02	1.00	1.01	0.98	0.90	0.85	0.82
Output	1.50	1.68	1.70	1.62	1.41	1.24	1.13
Panel B. Without equity issuance ($\chi = 0$)							
Debt to output (modern)			1.59	0.85	0.34	−0.13	−0.60
Interest rate			0.047	0.047	0.047	0.047	0.047
Fraction constrained			0	0.50	0.68	0.77	0.83
Capital to output (modern)			2.65	2.42	2.27	2.15	2.05
TFP (modern)			1.003	0.918	0.878	0.849	0.827
Loss misallocation, percent			0.0	2.7	3.8	4.4	4.7
Fraction producers modern			0.93	0.61	0.49	0.41	0.35
Fraction output modern			0.99	0.88	0.78	0.67	0.58
Consumption			1.01	0.91	0.87	0.84	0.82
Output			1.70	1.42	1.29	1.20	1.13

TABLE 4—AGGREGATE IMPLICATIONS OF FINANCE FRICTIONS: CLOSED ECONOMY

	$\theta = 1$	$\theta = 0.75$	$\theta = 0.50$	$\theta = 0.25$	$\theta = 0$
Panel A. With equity issuance ($\chi = 0.10$)					
Debt to output (modern)	1.21	1.07	0.84	0.58	0
Equity to output (modern)	0.30	0.32	0.43	0.58	0
Interest rate	0.049	0.042	0.033	0.019	−0.060
Fraction constrained	0	0.42	0.77	0.94	1.00
Capital to output (modern)	2.60	2.57	2.49	2.42	2.18
TFP (modern)	1.003	0.990	0.957	0.925	0.766
Loss misallocation, percent	0.0	1.2	4.7	7.3	9.8
Fraction producers modern	0.93	0.93	0.93	0.87	0.29
Fraction output modern	0.99	0.99	0.98	0.96	0.45
Consumption	1.00	0.99	0.94	0.91	0.71
Output	1.69	1.65	1.55	1.46	1.06
Panel B. Without equity issuance ($\chi = 0$)					
Debt to output (modern)	1.51	1.49	1.51	0.80	0
Interest rate	0.049	0.031	0.010	−0.060	−0.060
Fraction constrained	0	0.59	0.93	1.00	1.00
Capital to output (modern)	2.60	2.76	2.83	2.87	2.18
TFP (modern)	1.003	0.902	0.841	0.782	0.766
Loss misallocation, percent	0.0	3.7	7.2	10.3	9.8
Fraction producers modern	0.93	0.58	0.46	0.33	0.29
Fraction output modern	0.99	0.90	0.80	0.64	0.45
Consumption	1.00	0.91	0.87	0.78	0.71
Output	1.69	1.47	1.33	1.19	1.06

9.7 Table 5 (productivity gap): finance costs are larger when modern tech is much better

A bigger modern–traditional productivity gap raises the cost of preventing entry into modern technology.

9.8 Table 6 (adoption and exit): finance matters via adoption/operation

- **Adoption columns:** with no finance, the share adopting productive technology drops from ≈ 0.86 **to** ≈ 0.18, and modern-sector TFP drops by about ≈ 0.26 **log points**.
- **Exit columns:** finance frictions lower the fraction of modern producers operating (from ≈ 0.67 **to** ≈ 0.44) and raise misallocation losses somewhat (but still far below the largest empirical estimates).

Table 5—Role of Productivity Gap: Open Economy

	$(1-\eta)\phi_u = 0$		$(1-\eta)\phi_u = 0.2$		$(1-\eta)\phi_u = 0.4$	
	"Korea"	No finance	"Korea"	No finance	"Korea"	No finance
Debt to output (modern)	1.5	−1.6	1.2	−0.6	1.1	−0.3
Equity to output (modern)	0.3	0	0.3	0	0.3	0
Consumption	0.77	0.77	1	0.82	1.32	0.92
Output	1.21	0.93	1.68	1.13	2.29	1.35
TFP (modern)	0.80	0.61	1	0.83	1.22	0.94
Loss misallocation, percent	0.1	0.4	0.3	4.7	0.2	9.0
TFP (traditional)	0.66	0.81	0.56	0.78	0.56	0.79
Fraction producers modern	0.78	0.13	0.93	0.35	0.93	0.28
Fraction output modern	0.84	0.16	0.99	0.58	1.00	0.77
Fraction employment modern	0.77	0.11	0.99	0.48	1.00	0.69

Table 6—Extensions of the Benchmark Model

	"Korea"	No finance	"Korea"	No finance	"Korea"	No finance
Debt to output (modern)	1.2	−0.6	1.2	−0.2	1.2	−0.6
Equity to output (modern)	0.3	0	0.3	0	0.3	0
Consumption	1.00	0.82	1.41	0.97	0.86	0.79
Output	1.68	1.13	2.41	1.47	1.30	1.08
TFP modern sector	1	0.83	1.29	0.99	0.91	0.79
Loss misallocation, percent	0.3	4.7	1.2	6.3	2.1	4.1
Fraction producers modern	0.93	0.35	0.93	0.39	0.36	0.25
Fraction productive producers modern	—	—	0.86	0.18	—	—
Fraction modern producers operating	1	1	1	1	0.67	0.44
SD output growth	0.58	0.32	0.58	0.26	0.59	0.40
Average product of capital, 1–5 versus 11+	0.08	0.73	0.27	0.22	0.13	0.19
Relative output growth, 1–5 versus 11+	0.09	0.12	0.28	0.08	0.10	0.08

Table 7—Economy without Entry

		SD product shocks	SD output growth	Misallocation loss, percent	Measured TFP losses, percent
Data		0.83	0.59		23.5
Baseline	"Korea"	0.83	0.57	2.5	2.5
	No finance		0.38	3.4	3.4
Volatile productivity shocks	"Korea"	2.49	2.94	5.6	5.6
	No finance		2.03	20.3	20.3
Persistent productivity	"Korea"	0.83	0.69	2.8	2.8
	No finance		0.48	8.2	8.2
Predetermined capital	"Korea"	1.45	0.59	0.9	10.0
	No finance		0.61	0.8	9.9
Variable markups	"Korea"	1.84	0.59	0.3	2.7
	No finance		0.43	1.1	2.5
Low elasticity of substitutes	"Korea"	0.73	0.59	1.5	0.6
	No finance		0.23	3.2	1.2
Capital-specific shocks	"Korea"	0.62	0.59	0.3	7.1
	No finance		0.58	1.1	9.3

9.9 Table 7 (no entry + robustness): finance alone cannot hit large measured losses without unrealistic shocks

In data, measured TFP losses are about $\approx 23.5\%$. In the baseline no-entry model, measured losses are only a few percent. Only with extremely volatile idiosyncratic shocks do measured losses approach data, but then output growth volatility is counterfactually large.

9.10 Table 8 (heterogeneity): dispersion statistics can overstate true TFP losses

With heterogeneity in production parameters across sectors (e.g., different α or η), the "measured" loss based on dispersion in y_i/k_i can be much larger than the true welfare-relevant loss.

Table 8—Role of Heterogeneity

	Labor share		Span of control		Collateral constraint		Borrowing rates
	"Korea"	No finance	"Korea"	No finance	"Korea"	No finance	"China"
Sector 1	$\alpha_1 = 0.44$		$\eta_1 = 0.55$		$\theta_1 = 0.15$	$\theta_1 = 0$	$r_1 = 0.05$
Sector 2	$\alpha_2 = 0.66$		$\eta_2 = 0.85$		$\theta_2 = 0.58$	$\theta_2 = 0$	$r_2 = 0.10$
Sector 3	$\alpha_1 = 0.89$		$\eta_1 = 0.95$		$\theta_1 = 0.85$	$\theta_1 = 0$	$r_3 = 0.15$
Misallocation loss, percent	2.6	2.9	1.1	1.5	1.8	2.3	1.6
Sector 1 TFP losses, percent	2.7	3.2	1.0	2.0	2.2	2.3	0.0
Sector 2 TFP losses, percent	1.9	2.4	1.5	2.3	1.5	2.3	1.0
Sector 3 TFP losses, percent	0.8	1.0	1.1	1.2	0.7	2.3	2.2
Measured TFP losses, percent	8.2	8.2	1.9	2.5	1.8	2.3	1.6

Table 9—Cross-Country Evidence

	Benchmark model		Model w/ exit		Data		
	"Korea"	No finance	"Korea"	No finance	Korea	China	Colombia
Measured TFP							
Overall losses, percent	0.3	4.7	2.1	4.1	16.2	22.4	17.7
Due to age	0.1	3.7	0.1	0.7	0.2	0.3	2.7
If K fixed	1.3	1.3	5.4	5.4	2.4	2.9	1.9
Among 1–5	0.5	1.3	2.6	4.7	16.2	22.8	11.4
Among 11+	0.0	0.8	0.7	2.5	15.7	21.7	18.6
var e	0.27	1.10	0.35	0.30	0.24		
ρ_e	0.25	0.40	0.11	0.21	0.29		
σ_e	0.50	0.96	0.68	0.58	0.49		
avg(Y/K) 1–5 versus 11+	0.08	0.73	0.13	0.19	0.21	0.15	−0.25

9.11 Table 9 (cross-country + age decomposition): most dispersion is within age

In Korea/China/Colombia data, overall measured losses are large (e.g., $\approx 22.4\%$ in China), but the between-age component is small. This pattern is hard to reconcile with a story where only young firms are constrained by finance.

9.12 Figures 3–4 (Korean crisis): dynamics

- Figure 3 shows model dynamics after a credit tightening: modern activity contracts, TFP falls, and misallocation spikes temporarily.
- Figure 4 documents in the data that TFP fell and measured misallocation rose in 1997–1998 Korea, alongside a drop in the number of producers.

10 Short summary

- In this framework, finance frictions have *first-order* effects on entry/adoption/operation, but only *modest* effects on steady-state within-sector capital misallocation when shocks match plant-level data.

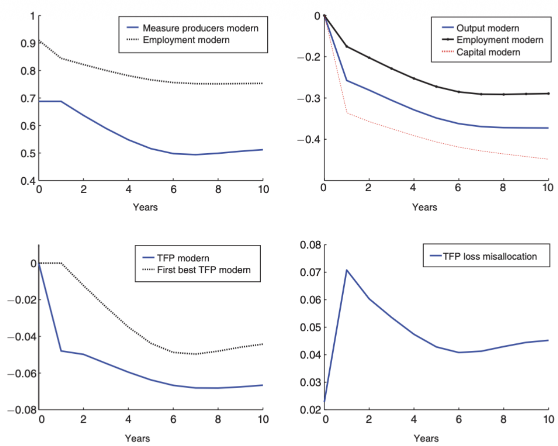

FIGURE 3. RESPONSE TO A CREDIT SHOCK: BENCHMARK MODEL

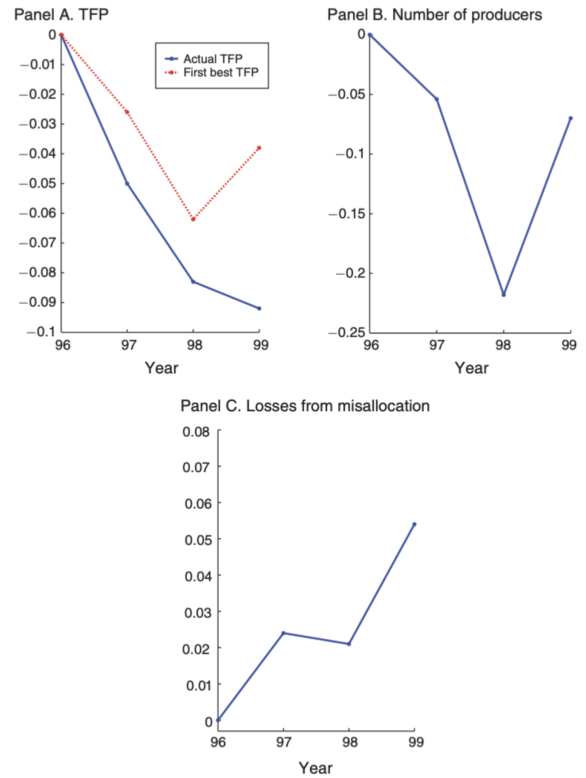

FIGURE 4. TFP DURING THE 1997–1998 KOREAN FINANCIAL CRISIS

- Therefore, large measured misallocation across countries likely reflects additional distortions beyond standard collateral-based finance frictions (or reflects that dispersion-based "measured" losses are not directly welfare-relevant when there is heterogeneity).

11 Conclusion

11.1 Summary

This paper evaluates whether **standard financial frictions**—modeled as collateral-based borrowing constraints (and limited equity issuance at entry)—can quantitatively explain the **large dispersion in marginal/average products of capital** observed in plant-level data and the large **TFP losses from misallocation** commonly inferred from that dispersion.

The paper's main conclusion is that **financial frictions alone generate only modest within-sector capital misallocation**, once the model is disciplined by plant-level moments (output growth volatility and persistence) and aggregate financing moments (debt ratios and equity issuance). In the calibrated model, constrained high-productivity plants **accumulate net worth and "grow out" of the constraint**, so the dispersion in the shadow cost of capital is limited in steady state. As a result, the implied **TFP losses from within-modern-sector capital misallocation are small** (a few percent), far below the large "measured" losses implied by dispersion-based accounting in the data.

At the same time, the paper shows that financial frictions can matter importantly through **extensive margins**—**entry into the modern technology**, **technology adoption**, and **operation/exit choices**—and through **crisis dynamics** (temporary spikes in measured misallocation during credit crunches). Thus, the welfare-relevant effects of finance are more plausibly tied to *which firms operate and which technologies they use*, not primarily to persistent within-sector capital wedges.

11.2 Key quantitative conclusions (portable anchors)

1. **Baseline (open economy, Korea calibration): small steady-state misallocation loss.** In Table 3 (Panel A), the benchmark economy features a misallocation loss of about **0.3%**.

2. **Eliminating external finance increases misallocation only modestly, but shrinks the modern sector sharply.** In Table 3 (Panel A), setting the collateral parameter to $\theta = 0$ raises the misallocation loss to about **4.7%**, but the *fraction of producers in the modern sector* collapses (from about **0.93** to about **0.35**), and aggregate output and consumption fall substantially. *Interpretation:* the large macro effect comes mainly from **less adoption/entry**, not from huge within-sector reallocation gains.

3. **Economy without entry: finance frictions still cannot match the large dispersion-based losses.** In Table 7, the data imply measured TFP losses of about **23.5%**, but the baseline no-entry model implies misallocation losses of only about **2.5%** (and about **3.4%** with no finance).

4. **To generate very large losses via finance, one needs unrealistically large idiosyncratic shocks.** In Table 7, volatile productivity shocks combined with no finance can generate very large losses (e.g., on the order of $\sim 20\%$), but this comes with counterfactually high volatility of plant-level output growth, violating the moments used to discipline the model.

5. **Measurement matters: "measured" dispersion can overstate true misallocation when technology/markup heterogeneity exists.** Table 8 illustrates that allowing heterogeneity in production parameters or borrowing rates can create a wedge between "measured TFP losses" and the welfare-relevant misallocation loss, reinforcing that dispersion-based statistics are not sufficient to pin down distortions.

11.3 Economic intuition: why finance generates only small within-sector misallocation

The mechanism is clearest from the firm's capital FOC under a borrowing constraint:

$$\mathrm{MRPK}_{i,t} = r + \delta + \mu_{i,t}, \qquad \mu_{i,t} \geq 0, \tag{1}$$

where $\mu_{i,t}$ is the shadow value of relaxing the collateral constraint. Misallocation in the modern sector is governed by the **dispersion in** $\mu_{i,t}$ across plants.

The key intuition is:

1. **Constraints bind mainly for low-net-worth (typically young/small) plants.** Constrained plants face $\mu_{i,t} > 0$, implying a higher shadow cost of capital and higher revenue products of capital.

2. **Productive constrained plants endogenously self-finance over time.** A high-productivity plant generates profits, accumulates net worth, and relaxes its own constraint. Therefore $\mu_{i,t}$ is **transitory** for plants that remain productive.

3. **Transitory wedges imply limited steady-state dispersion in marginal products.** With empirically realistic shock processes (matching plant output moments), the distribution of $\mu_{i,t}$ is not dispersed enough, for long enough, to generate large aggregate TFP losses.

4. **Large macro effects of finance therefore operate via extensive margins.** When finance is tight, fewer plants enter the modern sector (and fewer adopt superior technologies), and some plants may choose not to operate. These margins can deliver large output effects even if within-sector misallocation remains modest.

11.4 What to take away for research on misallocation

- **Dispersion in capital products is not a sufficient statistic for "finance-driven misallocation."** One must use dynamics (how fast firms grow out of constraints) to distinguish finance wedges from other sources.

- **If the goal is to explain large measured dispersion, finance frictions alone are unlikely to be enough.** Additional wedges (policy distortions, market power/markups, measurement error, or heterogeneous technologies) are needed, or the implied shock processes become inconsistent with micro moments.

- **A promising finance×misallocation agenda should emphasize entry/adoption and crisis states.** The model indicates that credit conditions can have first-order effects through selection into technologies and sectors, and through transitional dynamics when credit supply tightens.

11.5 Takeaway

Collateral-based financial frictions generate only small steady-state TFP losses from within-sector capital misallocation because productive firms quickly accumulate net worth and escape constraints; finance matters more through entry/adoption/operation margins and through crisis dynamics than through persistent dispersion in marginal products.